

D880

Evolutionary Biology

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A bone fragment found near an invented river canyon, now in a museum case with LED lighting, contains 25% of the carbon-14 expected in living tissue. The half-life of carbon-14 is 5,730 years. About how old is the bone?

- A. About 5,730 years
- B. About 4,298 years
- C. About 11,460 years
- D. About 17,190 years

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A sample is 17,190 years old. What percent of its original carbon-14 remains? (half-life 5,730 years)

- A. 33%
- B. 25%
- C. 0%
- D. 12.5%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A 100-amino-acid protein is compared across four invented species. Species Q differs from R by 3 amino acids, from S by 15, and from T by 27. Species T lives in the same forest as Q and looks most like it. Which species is Q's closest relative?

- A. T
- B. R
- C. T, because more differences mean a closer relationship
- D. S

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A gene accumulates differences between two lineages at about 2 differences per million years. Two species differ at 12 sites. About how long ago did they split?

- A.** It cannot be estimated from sequences.
- B.** About 6 million years
- C.** About 12 million years
- D.** About 24 million years

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

On an invented island studied by a field station with solar panels, a drought leaves only very small and very large seeds. Finches with medium beaks struggle to eat either, while small- and large-beaked finches do well. The station has recorded weather since 1990. Which type of selection is acting?

- A.** Directional selection
- B.** Genetic drift
- C.** Disruptive selection
- D.** Stabilizing selection

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Over many generations, a trait's average stays the same but its range narrows as extreme values become rare. Which type of selection does this show?

- A.** Stabilizing selection
- B.** Directional selection
- C.** Disruptive selection
- D.** No selection, since the average did not change

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Twenty people from a large mainland population settled an isolated island 300 years ago, traveling on a ship later sold for scrap. One settler carried a rare recessive allele. Today that allele is far more common on the island than on the mainland, and it brings no survival advantage. What best explains this?

- A. The founder effect
- B. Natural selection for the allele
- C. A mutation that happened on the island
- D. Gene flow from the mainland

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

In an invented bird species, males with longer, brighter tail feathers are caught by hawks more often. The species nests in a valley with many tall pines. Yet long, bright tails remain common. Why?

- A. Females prefer to mate with long-tailed males, so those males leave more offspring.
- B. The tall pines hide long-tailed birds.
- C. Long tails help males escape hawks.
- D. Every trait that persists must improve survival.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Some whale species, which can dive over 1,000 meters, have small pelvic bones buried in their bodies that do not connect to any leg. A museum displays one in a case with a touchscreen. How do biologists interpret these bones?

- A. As proof the bones evolved for deep diving
- B. As analogous structures unrelated to ancestry
- C. As vestigial structures showing whales descended from four-legged land ancestors
- D. As useless, so they tell us nothing about evolution

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

During an evolution lesson taught in a room with a new smartboard, a learner says: "Giraffes stretched their necks to reach leaves, so their babies were born with longer necks." The class had just watched a nature documentary. What is the best teaching response?

- A. Say individuals always evolve during their lifetimes to meet their needs.
- B. Tell the learner evolution is only a theory, so both views are fine.
- C. Agree, since stretching over many generations does lengthen necks.
- D. Ask how neck length varied among giraffes born in each generation, and which ones left more offspring, to shift the explanation to inherited variation and selection.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Which statement about evolution is accurate?

- A. Organisms develop new traits because they need them.
- B. A population's allele frequencies change over generations.
- C. An individual evolves when it adapts during its life.
- D. Evolution always produces more complex organisms.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

An invented wild cat species survives as one isolated population of about 50 animals in a reserve with a new visitor center. Its numbers have been steady for a decade, but genetic tests show very low diversity and signs of inbreeding. Which conservation action best addresses the genetic problem?

- A. Nothing; a population with steady numbers is always safe.
- B. Build a larger visitor center to raise funding.
- C. Breed more cats only from within the same group.
- D. Bring in individuals from other populations or build habitat corridors to restore gene flow.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A bone fragment found ~~near an invented river canyon, now in a museum case with LED lighting,~~ contains 25% of the carbon-14 expected in living tissue. The half-life of carbon-14 is 5,730 years. About how old is the bone?

Target word: age from isotope remaining **Decides:** 25% = two half-lives.

A	Dropped element	Counts one half-life; that leaves 50%.
B	Inverted operation	Takes 75% of one half-life, treating decay as linear.
C	Correct	100% → 50% → 25% is two half-lives: $2 \times 5,730$.
D	Confusion set	Three half-lives would leave 12.5%.

2. Answer D Concept 1 · transfer

A sample is 17,190 years old. What percent of its original carbon-14 remains? (half-life 5,730 years)

Target word: solve for percent remaining **Decides:** $17,190 / 5,730 = 3$ half-lives.

A	Inverted operation	Divides 100% by 3 instead of halving three times.
B	Wrong stem number	Counts two half-lives.
C	Absolute language	Decay never reaches exactly zero after a few half-lives.
D	Correct	$(\frac{1}{2})^3 = 1/8 = 12.5\%$.

3. Answer B Concept 2

A 100-amino-acid protein is compared across four invented species. Species Q differs from R by 3 amino acids, from S by 15, and from T by 27. ~~Species T lives in the same forest as Q and looks most like it.~~ Which species is Q's closest relative?

Target word: closest relative **Decides:** Fewest sequence differences.

A	Camouflage	Shared habitat and appearance do not outweigh 27 differences.
B	Correct	3 differences means the most recent common ancestor.
C	Inverted operation	More differences mean a more distant relationship.
D	Adjacent category	Moderately related, but R is closer.

4. Answer B Concept 2 · transfer

A gene accumulates differences between two lineages at about 2 differences per million years. Two species differ at 12 sites. About how long ago did they split?

Target word: solve for divergence time **Decides:** Time = differences / rate.

A	Absolute language	A calibrated clock gives a reasonable estimate.
B	Correct	$12 / 2 = 6$.
C	Dropped element	Ignores the rate.
D	Inverted operation	Multiplies instead of dividing.

5. Answer C Concept 3

On an invented island ~~studied by a field station with solar panels~~, a drought leaves only very small and very large seeds. Finches with medium beaks struggle to eat either, while small- and large-beaked finches do well. ~~The station has recorded weather since 1990.~~ Which type of selection is acting?

Target word: selection type **Decides:** Both extremes are favored.

A	Confusion set	Directional favors only one extreme.
B	Adjacent category	Survival here depends on beak size, not chance.
C	Correct	Extremes succeed and the middle is selected against.
D	Inverted operation	Stabilizing favors the middle.

6. Answer A Concept 3 • transfer

Over many generations, a trait's average stays the same but its range narrows as extreme values become rare. Which type of selection does this show?

Target word: name the selection from the pattern **Decides:** Same mean, less variation.

A	Correct	Middle values are favored and extremes removed.
B	Confusion set	Directional shifts the mean.
C	Inverted operation	Disruptive widens variation toward both extremes.
D	Dropped element	Ignores the change in variation.

7. Answer A Concept 4

Twenty people from a large mainland population settled an isolated island 300 years ago, ~~traveling on a ship later sold for scrap~~. One settler carried a rare recessive allele. Today that allele is far more common on the island than on the mainland, and it brings no survival advantage. What best explains this?

Target word: cause of the frequency difference **Decides:** A small founding group sampled alleles by chance.

A	Correct	The small starting group over-represented the allele by chance.
B	Confusion set	The stem says it brings no advantage.
C	Adjacent category	The allele arrived with a settler.
D	Inverted operation	Gene flow would make the island more like the mainland.

8. Answer A Concept 5

In an invented bird species, males with longer, brighter tail feathers are caught by hawks more often. ~~The species nests in a valley with many tall pines.~~ Yet long, bright tails remain common. Why?

Target word: why a costly trait persists **Decides:** The trait raises mating success.

A	Correct	Sexual selection can outweigh survival costs.
B	Camouflage	The habitat detail does not explain mating success.
C	Inverted operation	The stem says long tails increase hawk capture.
D	Absolute language	Reproductive success, not survival alone, drives fitness.

9. Answer C Concept 6

Some whale species, ~~which can dive over 1,000 meters,~~ have small pelvic bones buried in their bodies that do not connect to any leg. ~~A museum displays one in a case with a touchscreen.~~ How do biologists interpret these bones?

Target word: meaning of a vestigial structure **Decides:** Reduced remnants of ancestral hind limbs.

A	Camouflage	Diving depth does not explain an unattached pelvis.
B	Confusion set	Analogous structures arise independently; these are inherited remnants.
C	Correct	Reduced structures reflect ancestry.
D	Absolute language	Even reduced structures are strong evidence of common descent.

10. Answer D Concept 7

During an evolution lesson ~~taught in a room with a new smartboard,~~ a learner says: "Giraffes stretched their necks to reach leaves, so their babies were born with longer necks." ~~The class had just watched a nature documentary.~~ What is the best teaching response?

Target word: correcting a Lamarckian explanation **Decides:** Acquired traits are not inherited; variation plus selection changes populations.

A	Absolute language	Individuals do not evolve; populations do.
B	Adjacent category	Misrepresents what a scientific theory is.
C	Confusion set	Endorses inheritance of acquired traits.
D	Correct	Targets the misconception with the correct mechanism.

11. Answer B Concept 7 • transfer

Which statement about evolution is accurate?

Target word: identify the correct description **Decides:** Evolution is change in allele frequencies in populations.

A	Confusion set	Need does not create heritable variation.
B	Correct	This is the definition of evolution.
C	Inverted operation	Individuals acclimate; populations evolve.
D	Absolute language	Evolution has no set direction toward complexity.

12. Answer D Concept 8

An invented wild cat species survives as one isolated population of about 50 animals ~~in a reserve with a new visitor center.~~ Its numbers have been steady for a decade, but genetic tests show very low diversity and signs of inbreeding. Which conservation action best addresses the genetic problem?

Target word: restoring genetic diversity **Decides:** Low diversity needs new alleles through gene flow.

A	Absolute language	Stable numbers can hide genetic decline.
B	Camouflage	Funding helps indirectly but does not fix low genetic diversity.
C	Dropped element	Adds numbers but not new alleles.
D	Correct	New alleles counter drift and inbreeding.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	7	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Radiometric dating	After each half-life, half of the parent isotope remains; age = number of half-lives × half-life.	Subtracting a fixed amount per half-life or counting one half-life too many.	A percent of isotope remaining and a half-life.
2	Molecular evidence of relatedness	Fewer differences in DNA or protein sequences mean a more recent common ancestor; a steady rate of change lets differences act as a molecular clock.	Linking relatedness to shared habitat or looks instead of sequence data.	A table of sequence differences between species.
3	Types of natural selection	Directional favors one extreme and shifts the mean; stabilizing favors the middle and narrows variation; disruptive favors both extremes.	Calling any selection that changes a trait directional.	Which trait values survive or reproduce best.
4	Founder effect	A few individuals starting a new population carry only a sample of alleles, so allele frequencies can differ from the source by chance.	Explaining a founder population's unusual allele frequency as adaptation.	A small group colonizes an isolated place.
5	Sexual selection	Traits that raise mating success can spread even if they reduce survival.	Assuming every trait that persists must improve survival.	A showy trait that attracts predators but also mates.
6	Anatomical and developmental evidence	Vestigial structures and shared embryonic features are traces of common ancestry.	Treating a reduced structure as proof it was designed for a current job, or as unrelated to ancestry.	A structure with little or no current function.
7	Populations evolve, not individuals	Evolution is a change in a population's allele frequencies over generations; traits acquired by use or need are not inherited.	Explaining evolution as organisms changing because they need to (Lamarckian thinking).	A learner explanation using "need," "tried," or "stretched."
8	Genetic diversity and conservation	Small, isolated populations lose genetic diversity through drift and inbreeding; restoring gene flow helps them adapt and recover.	Thinking a population is safe once its numbers stabilize.	A small, isolated population with low genetic variation.

Course scope (public description): Evolutionary Biology provides a comprehensive exploration of key concepts and mechanisms of evolution. Students will learn about the evidence supporting evolution, including fossils and molecular data, the mechanisms driving evolutionary change like natural selection and genetic drift, and the historical and contemporary factors influencing the diversity of life. The course also addresses the importance of biodiversity conservation and equips educators with strategies to effectively teach evolutionary concepts in the classroom, emphasizing evidence-based applications and critical thinking.